Adding two arrays

Why [1, 2, 3] + [4, 5, 6] is a string

array-plus.js

```
1  console.log([1, 2, 3] + [4, 5, 6]); // "1,2,34,5,6"
```

output `"1,2,34,5,6"`

Why it happens

01 + has no array mode
It only adds numbers or joins strings.

02 Arrays become strings
JS calls toString(): [1, 2, 3] becomes "1,2,3".

03 Strings get joined
"1,2,3" + "4,5,6" glues the 3 and the 4 together.

04 Result is a string
typeof the result is "string", not an array.

What JavaScript does

what-js-does.js

```
1  // step 1: [1, 2, 3] -> "1,2,3" (toString)
2  // step 2: [4, 5, 6] -> "4,5,6"
3  // step 3: "1,2,3" + "4,5,6" -> "1,2,34,5,6"
```

@specsycoder

Guess the output

More array + coercion questions

merge.js

```
1  const a = [1, 2, 3];
2  const b = [4, 5, 6];
3  [...a, ...b]; // [1, 2, 3, 4, 5, 6]
4  a.concat(b); // [1, 2, 3, 4, 5, 6]
```

expression	result	why
[] + []	""	"" + ""
[] + {}	"[object Object]"	"" + "[object Object]"
[1] + 1	"11"	"1" + 1 joins
[2] * [3]	6	* converts both to numbers
[1, 2] + 3	"1,23"	"1,2" + 3

Traps to mention

- **push does not merge.** a.push(b) adds the whole array as ONE item.
- **Spread and concat are shallow.** Nested objects are still shared.

Your 20-second answer

"The + operator does not work on arrays, so JavaScript converts both to strings with toString. [1, 2, 3] becomes '1,2,3', [4, 5, 6] becomes '4,5,6', and + joins them into '1,2,34,5,6'. To merge arrays I use the spread operator or concat."